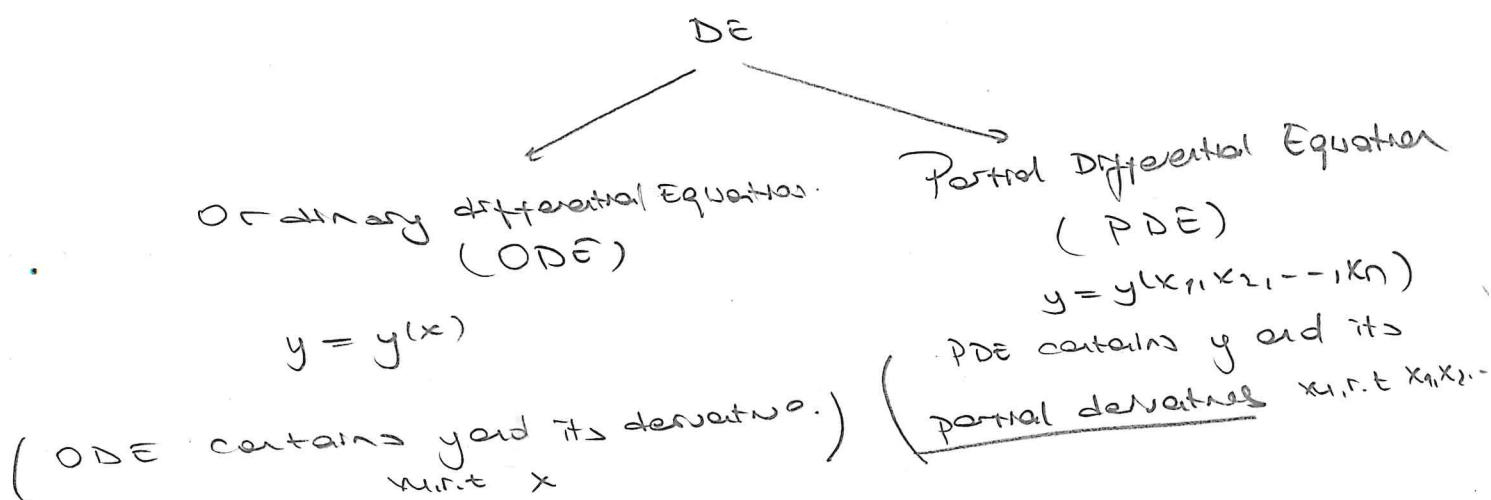


DIFFERENTIAL EQUATIONS LECTURE NOTES

WEEK I

Definition A differential equation (DE) is an equation involving an unknown function (dependent variable) and independent variables, e.g. $y' = y(x)$ y dep, x indep; $y = y(t)$ y dep, t indep.

→ we have two types of DEs.



Ex ① $y' = 4x + 5$ $y = y(x)$ (ODE) x indep, y dep.

② $y'' + yy' + 4ty = 0$ (ODE) t indep, y dep.

③ $\left(\frac{d^2u}{dx^2}\right)^3 - x \frac{du}{dx} + xu = 0$ (ODE) u : dependent, x indep.

④ $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ or $u_t = u_{xx}$ (Heat eq) (PDE) u : dependent $u = u(x, t)$
 t, x : indep.

⑤ $u_{xx} + u_y = 0$ (Laplace Eq) (PDE) u : dep.
 x, y : indep.

⑥ $\begin{cases} \frac{dx}{dt} = 3x + 5y \\ \frac{dy}{dt} = 4x - 2y \end{cases}$ System of ODE

⑦ $\frac{d}{dx}(e^{ax}) = a \cdot e^{ax}$ NOT A DIF EQ
IT IS AN IDENTITY
 $\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \frac{du}{dx}$ DIF IDENTITY

(1)

Definition The highest derivative of the unknown function appearing in DE is called the order of DE.

Ex 7 ① $y''' + 2y' + 3ty = 0$ ODE of order three
third order DE.

② $y'' + (y')^4 + 3ty = 0$ 2nd order DE

Power of first order derivative

Definition The degree of a differential equation is the power of the highest order derivative in the equation.

Ex 7 ① $\frac{dy}{dt} + y^2 = 1$ ODE, 1st order, 1st degree

② $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} + u = e^t$ PDE, 2nd order, 1st degree.

③ $\left(\frac{d^2 y}{dx^2}\right)^3 + \frac{dy}{dx} - 3xy^2 = \sin x$ ODE, 2nd order, 3rd degree.

④ $\frac{\partial^3 u}{\partial x^3} + \frac{\partial u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ PDE, 3rd order, 1st degree

⑤ $((y'')^2)^{2/3} = 1 + (y')^2 \Rightarrow (y'')^4 = (1 + (y')^2)^3$ 2nd order
4th degree.
Write like a polynomial.

Definition The general form of an nth order ODE can be written as

$$F(t, y, y', \dots, y^{(n)}) = 0$$

Ex 7 $y'' + 2ty + 4t = 0$ $F(t, y, y', y'') = 0$

→ If the $y^{(n)}$ can be left alone, then one can obtain the standard form of an nth order ODE. That is;

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)})$$

Ex 7 $y' = f(t, y)$, $y' = 3t^2 + \sin y \dots$ first order DE.
 $y'' = f(t, y, y')$, $y'' = 3t^2 y + (y')^2 \dots$ second order DE.
 $y''' = f(t, y, y', y'')$, $y''' = \cos t + e^t \cdot y' + (y'')^2$ third order DE.
 $\sin^2 t \cdot y''' + (t y'')^2 + e^t y = 0 = F(t, y, y', y'', y''')$

Why Differential Equations are Interesting

- ① Has applications in real life problems. The dependent and independent variables often represent certain physical quantities such as volume, price, velocity, length, time etc. Applications are very diverse, spanning essentially all branches of science and technology.
- ② Understanding and testing how a function changes with an independent variable is easier than writing a direct formula. For example, consider $y(t)$ is given at a special time point t_0 of t , if we know how fast $y(t)$ change we may reconstruct $y(t)$ from this information.

SOLUTION OF DIFFERENTIAL EQUATIONS

Definition ① Suppose we have

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)}) \quad \text{①}$$

The given conditions

$$y(t_0) = y_0, y'(t_0) = y_1, \dots, y^{(n-1)}(t_0) = y_{n-1} \quad \text{②}$$

are called initial conditions.

- ② The problem containing ① + ② is called an initial value problem (IVP)

Ex $y' = f(t, y)$ } I.V.P.
 $y(t_0) = y_0$

$y'' = f(t, y, y')$
 $y(t_0) = y_0, y'(t_0) = y_1$ } I.V.P.

Definition: (Solution of a DE)

$$y^{(n)} = f(t, y, y', \dots, y^{(n-1)})$$

$$y(t_0) = y_0, y'(t_0) = y_0', \dots, y^{(n-1)}(t_0) = y_{n-1}$$

} I.V.P

①

②

① A function $y = \phi(t)$ that satisfies the DE ① is called a solution of DE ①

② A function $y = \phi(t)$ that satisfies both ① and ② is called a solution of I.V.P.

③ A solution containing constants is called a general solution of DE ①

④ Solution of the form $y = \phi(t)$ is called explicit solution
Solution of the form $f(y, t) = 0$ is called implicit solution

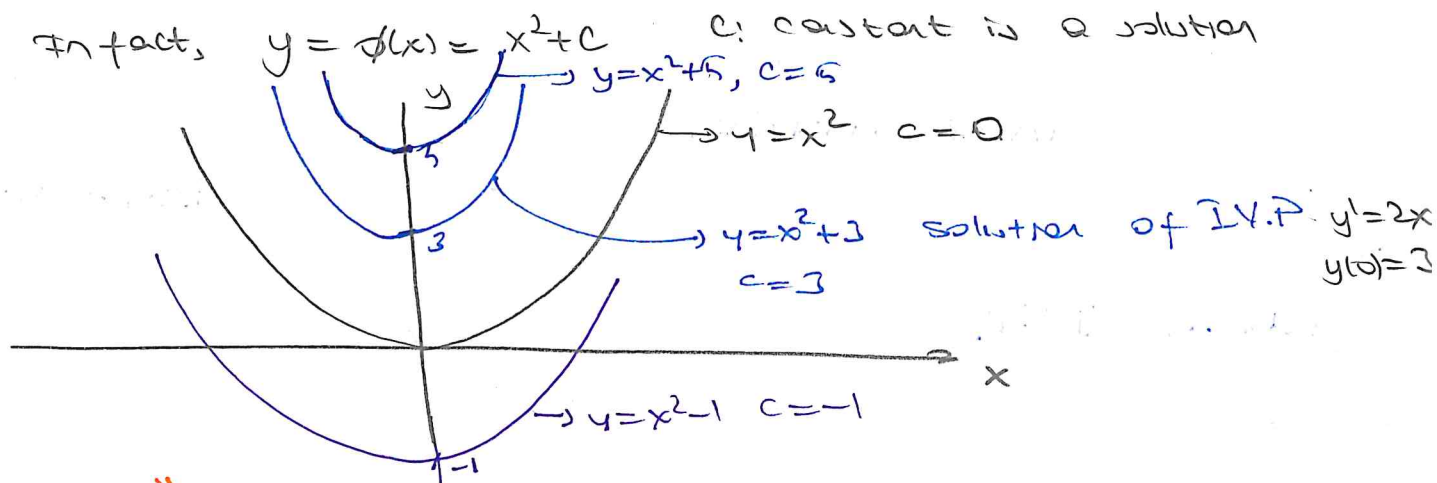
Ex ① $y' = 2x$

• $y = \phi_1(x) = x^2 \Rightarrow$ a solution for $y' = 2x$ since

$$\phi_1'(x) = 2x$$

• $y = \phi_2(x) = x^2 + 5$ is also a solution of DE

In fact, $y = \phi(x) = x^2 + C$ C: constant is a solution



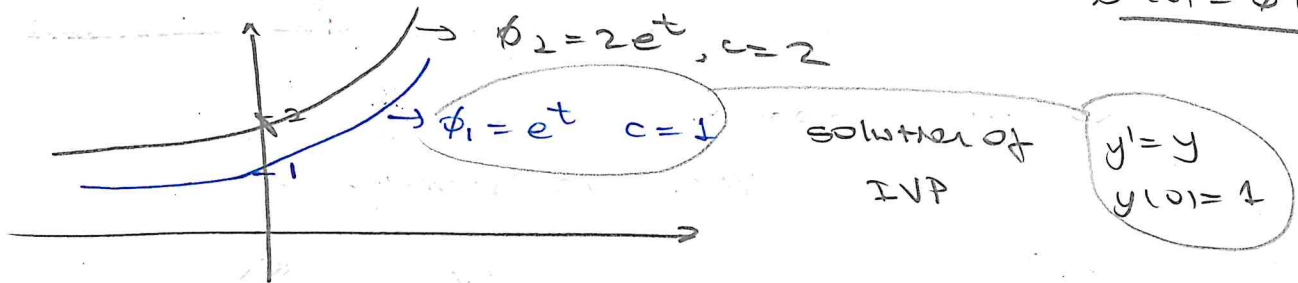
"solutions (integral) curves for the D.E."

(1.4)

Ex $y' = 2x$ } IVP $y = \phi(x) = x^2 + C$
 $y(0) = 3$ Put initial value $y = 3, x = 0 \Rightarrow 3 = 0^2 + C$
 $C = 3$

Then, the solution of IVP $y = \phi(x) = x^2 + 3$

Ex $y' = y$ $y = y(t)$
 $\phi(t) = C \cdot e^t$ is a solution since $\phi'(t) = C \cdot e^t = \phi(t)$
 $\phi'(t) = \phi(t)$



Ex $y'' + y = 0 \Rightarrow y'' = -y$
 $\phi_1(x) = \cos x$
 $\phi_2(x) = \sin x$ are both solutions of $y'' = -y$
 $(\cos x)'' = -\cos x$
 $(\sin x)'' = -\sin x$

$\phi_3(x) = c_1 \sin x + c_2 \cos x$ it is also a solution of $y'' + y = 0$

general solution

Remark ORDER OF DE = NUMBER OF CONSTANTS IN GENERAL EQUATION

Ex $g(x, y) = x^2 + y^2 - 25 = 0$ function is the implicit solution
of $x + yy' = 0$ $y' = \frac{dy}{dx}$

$\frac{\partial g}{\partial x} = 2x + 2yy' = 0 \Rightarrow y' = -x/y$ Insert equation $x + yy' = 0$
 $\Rightarrow x + y(-x/y) = 0$

Ex $y = 2 \sin x + 3 \cos x$ is the explicit solution of $y'' + y = 0$
Since, $y' = 2 \cos x - 3 \sin x$ $y'' + y = 0$
 $y'' = -2 \sin x - 3 \cos x$ $(-2 \sin x - 3 \cos x) + (2 \sin x + 3 \cos x) = 0$

(5)

ex $\phi(t) = e^{2t}$ is an explicit solution of $y' = 2y$ since $\phi' = 2e^{2t} = 2\phi$
Interval of solution (the points where $\phi(t)$ is defined) = $\mathbb{R}$

ex $\phi(t) = t^2 - \frac{1}{t}$ is an explicit solution of $t^2 y'' - 2y = 0$

$$\left[\phi'(t) = 2t + \frac{1}{t^2} \quad \phi''(t) = 2 - \frac{2}{t^3} \quad t^2 \left(2t + \frac{1}{t^2} \right) - 2 \left(t^2 - \frac{1}{t} \right) = 0 \right]$$

Interval of solution: $\mathbb{R} \setminus \{0\}$

ex $x^2 + y^2 = 25$ is implicit solution of $x + yy' = 0$

We need to check both $y = \pm \sqrt{25 - x^2}$ $y = -\sqrt{25 - x^2}$

are solutions.

$$y = (25 - x^2)^{1/2} \quad \frac{dy}{dx} = \frac{1}{2} (25 - x^2)^{-1/2} \cdot (-2x) = -\frac{x}{\sqrt{25 - x^2}}$$

$$\text{insert to equation } x + (25 - x^2)^{1/2} \frac{-x}{(25 - x^2)^{1/2}} = 0$$

Similarly it can be seen $y = -\sqrt{25 - x^2}$ is also a solution.

Interval of solution $25 - x^2 > 0 \Rightarrow x \in (-5, 5)$

Exercise Show that $4x^2 - y^2 = 0$ is an implicit solution
of $yy' - 4x = 0$ and find interval of solution.

SOLUTION CURVES AND DIRECTION FIELDS

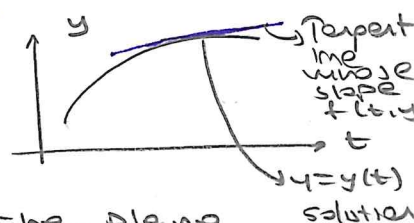
Question How can we say something about the solution without solving the D.E

Answer (1) By looking at the graph of solution (direction fields)
(2) Existence and Uniqueness Theorem

① SOLUTION CURVES AND DIRECTION FIELDS

→ Consider $y' = f(t, y)$

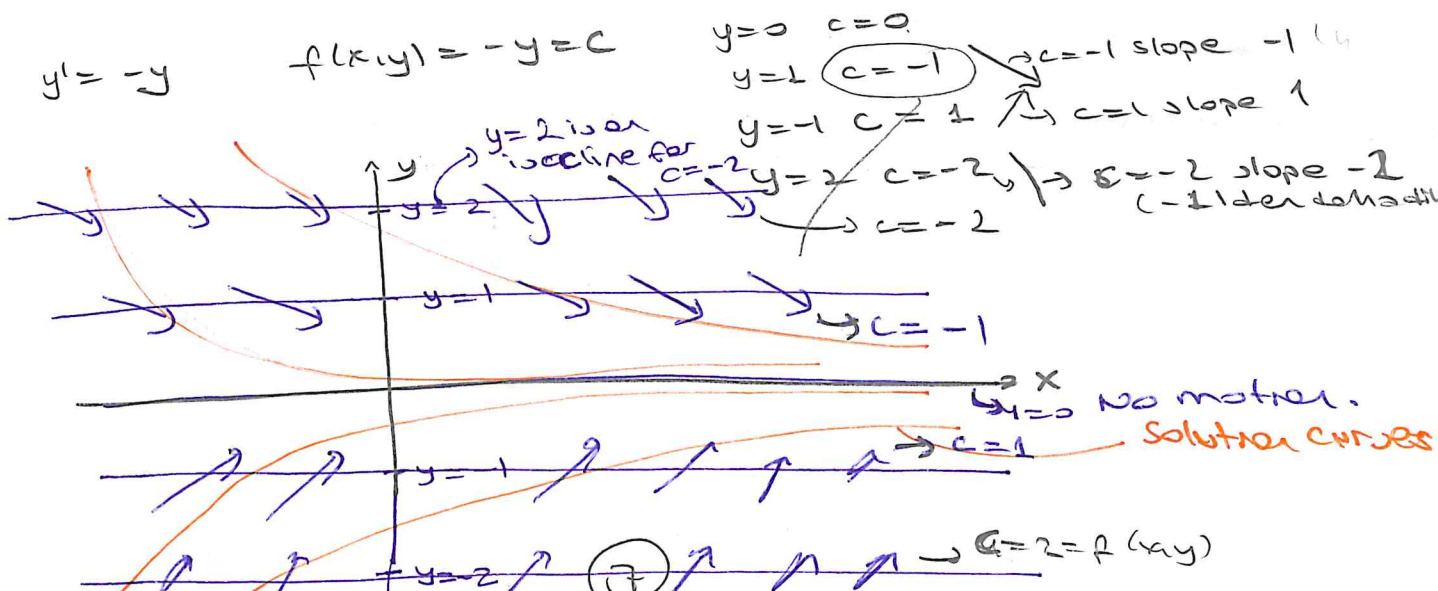
At each point (t, y) on the plane the value $f(t, y)$ equals slope of the tangent line to the solution curve passing through that point



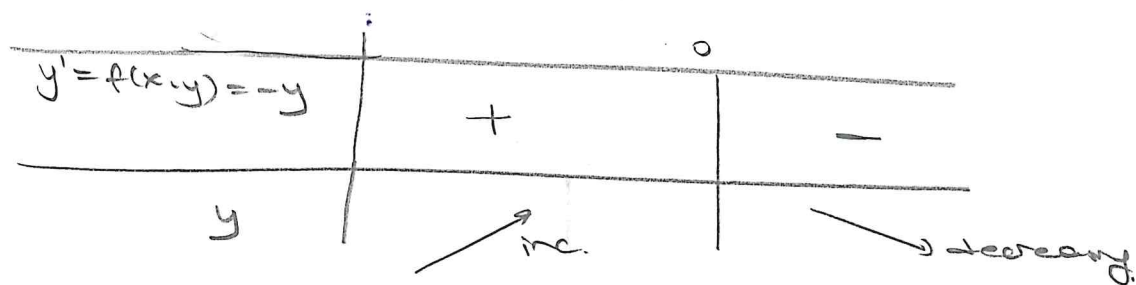
Ex $y' = -y$ (Analytical solution is $y = C \cdot e^{-x}$)

Draw direction field and solution (integral) curve without solving DE.

- ① Pick a slope $C \in \mathbb{R}$ (constant)
- ② Plot $f(x, y) = C$. (It is a level curve of $f(x, y)$ corresponding to level constant of C , which is called isoclines)
- ③ Draw small line segments

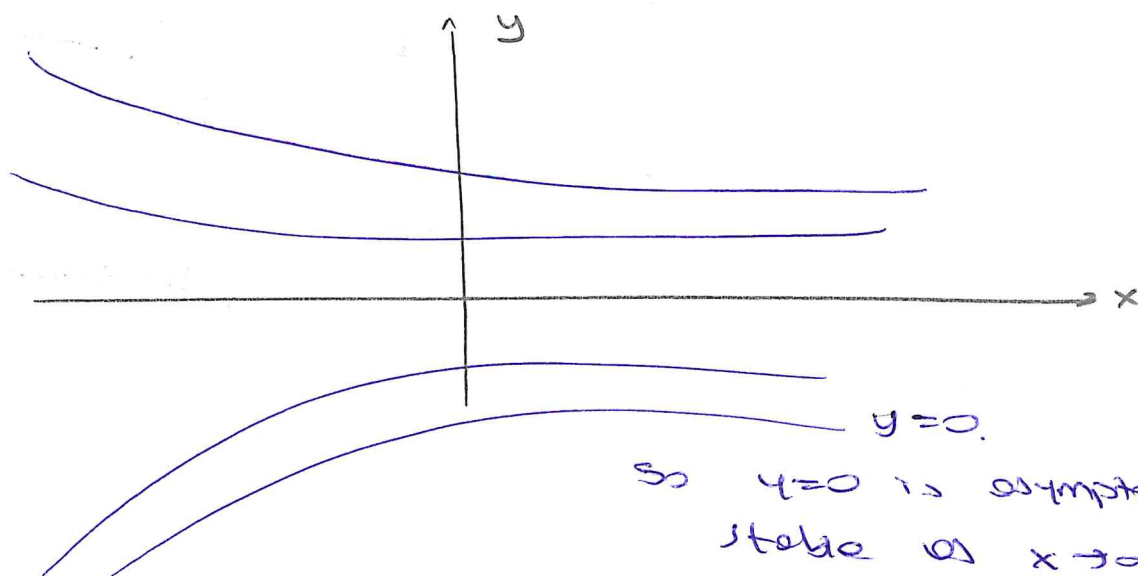


Behavior of solution



$y=0$ is equilibrium, since when $y=0$ $f(x, y)=0 \Rightarrow D.E.$

$$y' = 0 \Rightarrow y = \text{constant}$$



→ The method of direction fields is a graphical method for displaying the general shape and behavior of solutions $y' = f(t, y)$

→ It doesn't require solving the differential equation

$$y' = f(x, y)$$

$y' = f(t, y)$ $\xleftrightarrow{\text{From a geometric view}}$ Direction fields

$y = \phi(t)$ is solution $\longleftrightarrow$ Integral curve

Solution Curves (Integral Curves)

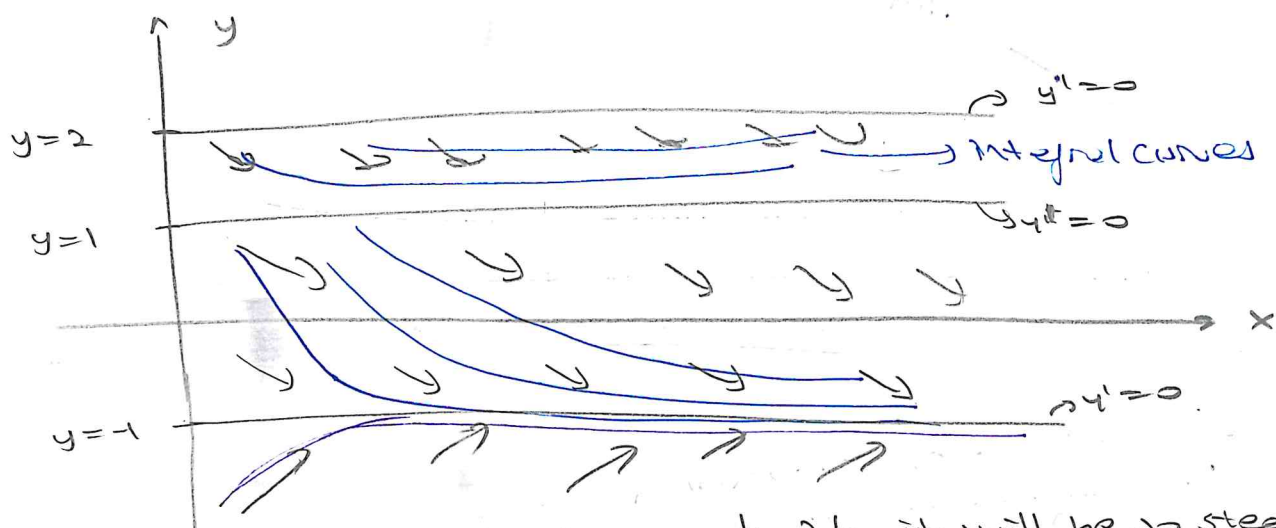
The set of solutions of $y' = f(t, y)$ is called family of solution curves or the set of integral curves

Ex $y' = (y^2 - y - 2)(1 - y)^2$ obtain direction field and

Solution curves (integral curves) $y = y(x)$

$$y' = 0 \quad (y-2)(y+1)(1-y)^2 = 0$$

$$y = 2 \quad y = -1 \quad y = 1 \quad \text{NO MOTION}$$



$y < -1$ let $y = -2$ then $y' = 36$ it will be so steep (down)

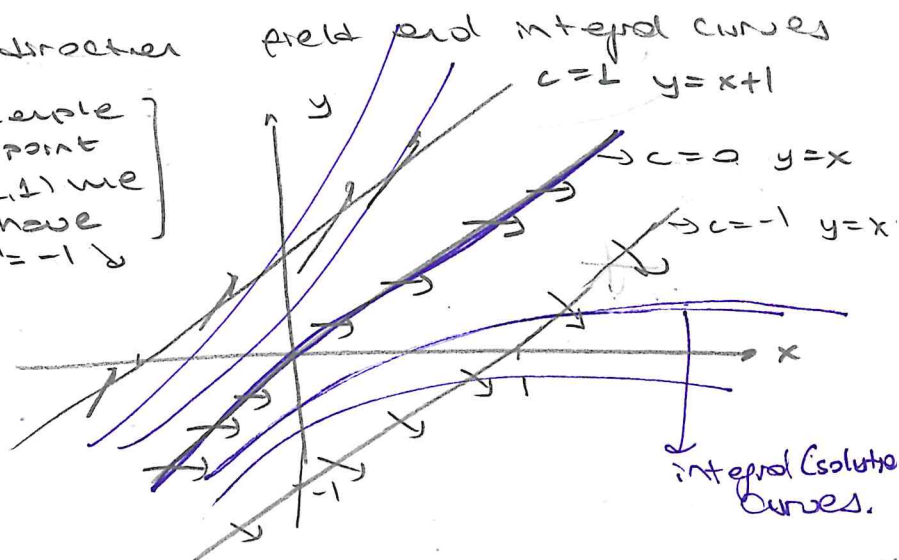
$-1 < y < 1$ let $y = 0$ then $y' = -2$

$2 < y < 1$ let $y = 1.5$ $y' = -0.3125$

Ex $y' = y - x$ obtain direction

$y' = c$	$y - x = c$	For example at point (2, 1) we have $y' = -1$
$c = -1$	$y = x - 1$	
$c = -2$	$y = x - 2$	
$c = -3$	$y = x - 3$	
$c = 0$	$y = x$	

$c = 0$ slope is 0 $\rightarrow$
 $c = 1$ slope is 1 $\nearrow$
 $c = -1$ slope is -1 $\searrow$

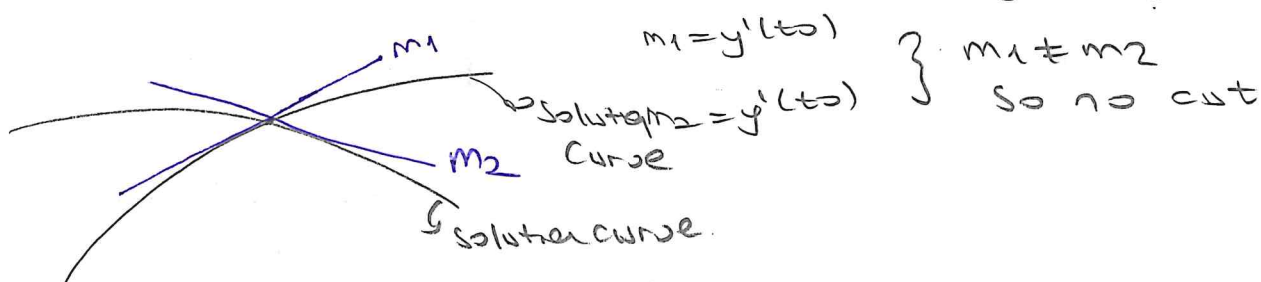


General solution $y = x + 1 + Ce^x$

Definition If all solutions approaches to equilibrium solution as $x \rightarrow \infty$ then equilibrium solution is called **asymptotically stable** otherwise, if all other solutions goes far away from equilibrium solution, then equilibrium solution is called **asymptotically unstable**.

Principle in drawing integral (solution) curves

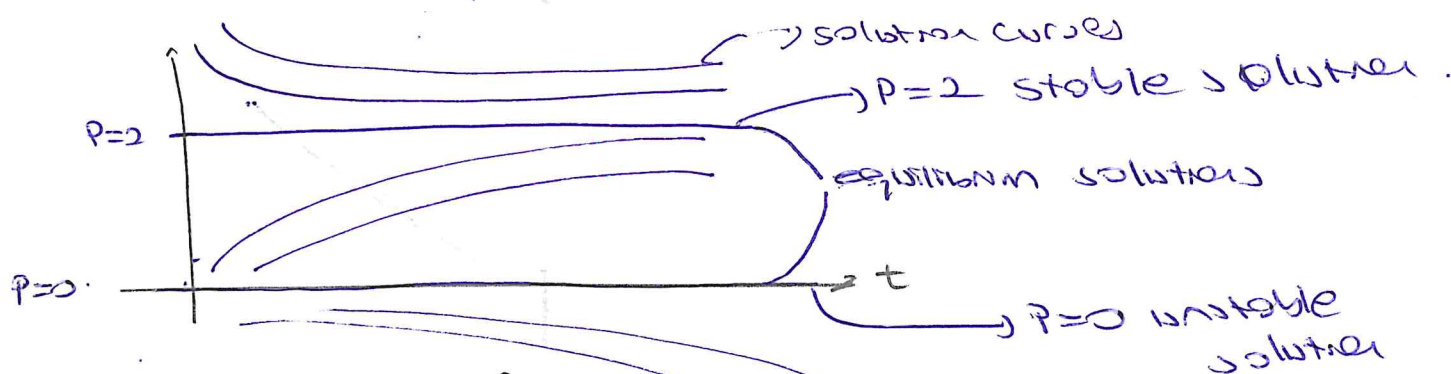
- ① Two solution curve CAN NOT CUT each other $y' = f(t, y)$
 $y' = f(t, y)$



- ② Two solution curve CAN NOT BE TANGENT
 NOT POSSIBLE!!
-

Ex $\frac{dP}{dt} = P(2-P)$ obtain direction field and integral curves

$P(2-P) = 0$ $P = 0$ $P = 2$ Equilibrium solutions



	0	2	
$P' = P(2-P)$	-	+	-
P			

Definition $y' = f(y)$ is called **AUTONOMOUS D.E.**

GENERAL SOLUTION OF THE ODE

A general solution of an ODE is a solution containing
arbitrary independent constants

PARTICULAR SOLUTION OF A.O. DDE

A particular solution of a ODE is derived from the general solution by setting the constants to particular values

SINGULAR SOLUTION OF AN ODE

A singular solution of an ODE is a solution that can't be obtained by assigning definite values to the arbitrary constants in the general solution.

$$\text{ex } \frac{dy}{dx} - e^x = 0 \quad \frac{dy}{dx} = e^x \quad dy = e^x dx$$

Solution $\int dy = \int e^x dx$ $y = e^x + C \rightarrow$ General solution
Arbitrary constant.

If $y(0) = 2$ condition is given, then $y(0) = e^0 + C = 2$
 $C = 1$ $y(x) = e^x + 1$ → particular solution

$$\text{ex } \frac{d^2 y}{dx^2} - e^x = 0 \quad \frac{d^2 y}{dx^2} = e^x \quad \frac{dy'}{dx} = e^x$$

$$\int dy' = \int e^x dx \quad y' = e^x + C_1 \quad y' = dy$$

$$\int dy = \int (e^x + C_1) dx \quad y(x) = e^x + C_1 x + C_2 \rightarrow \text{2 arbitrary constant}$$

$$7 \quad y - x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^2 = 0$$

$$y = cx - c^2 \rightarrow \text{General solution}$$

$$y = 2x - 4 \rightarrow \text{Particular solution}$$

(11) $y = \frac{x^2}{4}$ → simpler solution
cost we obtained from

FIRST ORDER FIRST DEGREE DIFFERENTIAL EQUATIONS

① SEPARABLE DIFFERENTIAL EQUATIONS

Definition A differential equation $y' = f(x, y)$ ① is called

separable if it can be expressed as $f(x, y) = p(x)q(y)$

→ In this case, ① becomes $y' = p(x)q(y)$ ②. Therefore,

we get

$$\frac{y'}{q(y)} = p(x) \quad \text{as long as } q(y) \neq 0$$

Remark Separation process may cause loss of some of solutions. If $q(y) = 0$ when $y = L$, then $y' = 0$ and hence ② is satisfied. This means that $y = L$ is a solution of ①

HOW TO SOLVE SEPARABLE DIFF. EQUATIONS?

① Separate the system as $\frac{dy}{q(y)} = p(x) dx$

② Integrate both sides

$$\int \frac{dy}{q(y)} = \int p(x) dx \quad \begin{aligned} \Theta'(y) &= \frac{1}{q(y)} \\ P'(x) &= p(x) \end{aligned}$$
$$\underline{\Theta(y) = P(x) + C} \quad \downarrow \text{General solution}$$

③ Check for $y = L$ if there is one by using $q(y) = 0$
 $q(L) = 0$. We have to check because we integrate

$\int \frac{dy}{q(y)}$ with $q(y) \neq 0$ and we are losing $q(y) = 0$

case. Therefore we must check if some solution

is lost or not

Ex 1 Solve $(x-2)y^3 dx + x^4(y-3)dy = 0$

$$\left(\frac{x-2}{y^3}\right)dx = \left(\frac{3-y}{y^3}\right)dy$$

$$\int \left(\frac{1}{x^3} - \frac{2}{x^4}\right)dx = \int \left(\frac{3}{y^3} - \frac{1}{y^2}\right)dy$$

$$-\frac{1}{2x^2} + \frac{2}{3x^3} + C_1 = -\frac{3}{2y^2} + \frac{1}{y} + C_2 \quad C_1, C_2 \text{ are arbitrary constants}$$

$$-\frac{1}{2x^2} + \frac{2}{3x^3} + \frac{3}{2y^2} - \frac{1}{y} = C \quad \text{general solution}$$

$C_2 - C_1$ arbitrary constant

$$\frac{dy}{dx} = -\frac{(x-2)}{x^4} \underbrace{\left(\frac{y^3}{y-3}\right)}_{q(y)}$$

$q(y)=0 \Rightarrow y=0$ is a solution that can be obtained from the general solution

2 $\frac{dy}{dx} = (x-2)^2(y-3)$

$$\frac{dy}{y-3} = \frac{dx}{(x-2)^2}$$

$$\ln|y-3| = \frac{(x-2)^3}{3} + C$$

$$|y-3| = e^{\frac{(x-2)^3}{3}} \cdot \underbrace{e^C}_{C > 0}$$

(A) $y = 3 \pm C e^{\frac{(x-2)^3}{3}} \quad C > 0$ General solution

(B) $q(y)=y-3 \quad q(y)=0 \Rightarrow y=3$ is a solution

(A) + (B) gives $y = 3 \pm C e^{\frac{(x-2)^3}{3}} \quad C \geq 0$

Same meaning with

$$y = 3 + C e^{\frac{(x-2)^3}{3}} \quad C \text{ arbitrary constant}$$

③ $y' = \frac{y-1}{x+3}$ $y(-1)=0$

$$\frac{dy}{dx} = \frac{y-1}{x+3}$$

$$\frac{dy}{y-1} = \frac{dx}{x+3}$$

General solution $\ln|y-1| = \ln|x+3| + \ln C$

$$|y-1| = |x+3| \cdot C$$

Particular solution $y(-1)=0$ $|-1-1| = |-1+3| \cdot C$

$$1 = 2C \quad C = 1/2$$

So, the particular solution $\rightarrow |y-1| = \frac{|x+3|}{2}$

④ If the question was only $y' = \frac{y-1}{x+3}$, $|y-1| = |x+3| \cdot C$ general solution
 $q(y) = y-1$ $q(y)=0$ $y=1$ is also obtained for $C=0$.

Exercises Solve following equations.

① $(1+y^2)dx + (1+x^2)dy = 0$

$$\arctan a + \arctan b = \arctan\left(\frac{a+b}{1-ab}\right)$$

$$\frac{dy}{dx} = -\frac{(1+y^2)}{1+x^2}$$

$$\frac{dx}{1+x^2} + \frac{dy}{1+y^2} = 0$$

$$\arctan x + \arctan y = C$$

$$\arctan\left(\frac{x+y}{1-xy}\right) = C \quad \tan C = \frac{x+y}{1-xy}$$

$q(y) = 1+y^2 \neq 0$ no solution comes from here.

② $y' = \frac{x-5}{y^2}$ $\frac{dy}{dx} = \frac{x-5}{y^2}$ $\int y^2 dy = \int (x-5) dx$

$$\frac{y^3}{3} = \frac{x^2}{2} - 5x + C \Rightarrow \text{general solution}$$

$$y = \left(3\left(\frac{x^2}{2} - 5x\right) + C\right)^{1/3}$$

$q(y) = \frac{1}{y^2} \neq 0$ no solution comes from here.

$$(3) \quad y' = \frac{3x^2 + 4x - 4}{2y - 4} \quad y(1) = 3$$

$$\int (2y - 4) dy = \int (3x^2 + 4x - 4) dx$$

$$y^2 - 4y = x^3 + 2x^2 - 4x + C \rightarrow \text{General solution}$$

$$y(1) = 3 \quad 3^2 - 4 \cdot 3 = 1^3 + 2 - 4 + C \quad C = -2$$

$$y^2 - 4y = x^3 + 2x^2 - 4x - 2 \rightarrow \text{Particular solution}$$

$$(4) \quad y' = \frac{xy^3}{1+x^2} \quad y(0) = -1$$

$$\frac{dy}{dx} = \frac{xy^3}{1+x^2}$$

$$\frac{dy}{y^3} = \frac{x dx}{1+x^2}$$

$$y^{-3} dy = \frac{x}{1+x^2} dx$$

$$\int y^{-3} dy = \int \frac{\overbrace{x dx}^{du/2}}{\underbrace{1+x^2}_u} \quad \begin{array}{l} 1+x^2 = u \\ 2x dx = du \end{array}$$

$$\int y^{-3} dy = \int \frac{1}{2u} du \Rightarrow -\frac{y^{-2}}{2} = \frac{1}{2} \ln u + C$$

$$y^{-2} = -2(1+x^2)^{1/2} - \frac{2C}{C}$$

$$y^{-2}(x) = -2(1+x^2)^{1/2} + C \rightarrow \text{General solution}$$

$$y(0) = -1 \quad 1 = -2(1+0)^{1/2} + C \quad C = 3$$

$$\boxed{y^{-2}(x) = -2(1+x^2)^{1/2} + 3}$$

$$(5) \quad y' = 6y^2 x \quad y(1) = 1/25$$

$$\int \frac{dy}{y^2} = \int 6x dx \Rightarrow -\frac{1}{y} = 3x^2 + C$$

$$y(1) = 1/25$$

$$-25 = 3 + C \quad C = -28$$

Particular solution

$$\boxed{-\frac{1}{y} = 3x^2 - 28}$$

$$(15)$$

$$\textcircled{6} \quad \frac{dy}{dt} = t \frac{1+y^2}{2+y^2} \Rightarrow \frac{2+y^2}{1+y^2} dy = t dt$$

$$\int \left(1 + \frac{1}{1+y^2}\right) dy = \int t dt \quad y + \arctan y = \frac{t^2}{2} + C \quad \text{general solution}$$

$$q(y) = \frac{1+y^2}{2+y^2} \neq 0 \quad \text{no solution}$$

$$\textcircled{7} \quad \frac{dy}{dx} = e^{-y} (2x-4) \quad y(5)=0$$

$$\int e^y dy = \int (2x-4) dx \quad \underbrace{e^y = x^2 - 4x + C}_{\text{general solution.}}$$

$$\underline{y(5)=0} \quad 1 = 25 - 20 + C$$

$$C = -4$$

$$\text{particular solution} \quad e^y = x^2 - 4x - 4$$

$$\textcircled{8} \quad 2x\sqrt{1-y^2} dx - dy = 0 \quad \int \frac{dy}{\sqrt{1-y^2}} = \int 2x dx$$

$$\arcsin y = x^2 + C \quad y(x) = \sin(x^2 + C) \rightarrow \text{general solution}$$

$\rightarrow$ In this example $|y| < 1$. so $y = \pm 1 \rightarrow$ also a solution

$$\textcircled{9} \quad y' \cot x + y = 2 \quad y(0) = -1$$

$$y' = (2-y) \tan x \quad \frac{dy}{2-y} = \frac{\sin x}{\cos x} dx$$

$$-\ln|2-y| = -\ln|\cos x| - \ln C = -\ln|C \cos x|$$

$$|2-y| = |C \cos x| \rightarrow \text{general solution}$$

$$q(y) = 2-y \quad q(y) = 0 \quad \text{for } y = 2 \rightarrow \text{a solution.}$$

(10) $xy dx - (x+2) dy = 0$ $\frac{dy}{dx} = \frac{xy}{x+2} = \underbrace{\frac{x}{x+2}}_{f(x,y)} \cdot \underbrace{y}_{g(y)}$

$\int \frac{dy}{y} = \int \frac{x}{x+2} dx \quad y \neq 0$

$\ln|y| = \int \left(1 - \frac{2}{x+2}\right) dx$

$\ln|y| = x - 2 \ln|x+2| + C_1$

$|y| = e^x \cdot \underbrace{e^{-2 \ln|x+2|}}_{\frac{1}{(x+2)^2}} \cdot \underbrace{e^{C_1}}_{C_2} =$

$|y| = C \cdot \frac{e^x}{(x+2)^2} \Rightarrow$ a parameter family of solution.

$g(y)=y \quad y=0$ is also a solution.

(11) $\frac{dy}{dx} = 2y^2 + xy^2$ } I.V.P $f(x,y) = \underbrace{(2+x)}_{p(x)} \underbrace{y^2}_{q(y)}$
 $y(-1) = 0$

$\int \frac{dy}{y^2} = \int (2+x) dx$

$-\frac{1}{y} = 2x + \frac{x^2}{2} + C$

$y = -\frac{1}{2x + \frac{x^2}{2} + C} \quad y \neq 0$

Put I.V.P $x=-1$
 $y=0$

$0 = \frac{1}{-2 + \frac{1}{2} + C}$

So there exists no such C

On the other hand: Check $y=0$ is satisfying D.E and also Initial condition

$y(-1) = 0$

So $y(x) \equiv 0$ is a solution for I.V.P.

(12) $y' = \frac{2 \cos 2x}{3+2y}$

separable

$y(0) = -1$

$y = -\frac{3}{2} + \sqrt{\sin 2x + 1/4}$

put $x=0, y=-1$. the solution with positive sign satisfies the I.C so the solution of I.V.P. is $y(x) = -\frac{3}{2} + \sqrt{\sin 2x + 1/4}$

$\int (3+2y) dy = \int 2 \cos 2x dx \Rightarrow 3y + y^2 = \frac{\sin 2x}{1} + C$

To find C_1 put $x=0, y=-1$
 $3y + y^2 = \sin 2x - 2$

$\frac{-2-C}{1} (y + \frac{3}{2})^2 - \frac{9}{4} = \sin 2x - 2$

II HOMOGENEOUS DIFF. EQUATIONS

Definition A function $f(x,y)$ is said to be homogeneous of degree n if $f(\alpha x, \alpha y) = \alpha^n f(x,y)$ for all values of x, y and α .

A first order differential equation $f(x,y)$ is said to be homogeneous type if $f(x,y)$ homogeneous of degree 0

$$f(\alpha x, \alpha y) = \alpha^0 f(x,y) = f(x,y)$$

for all values of x, y and α .

Ex $f(x,y) = \sqrt{x^2 + y^2}^3$ $f(\alpha x, \alpha y) = \sqrt{\alpha^2 x^2 + \alpha^2 y^2}^3 = \alpha^{3/2} \sqrt{x^2 + y^2}^3$
homogeneous of degree $3/2$. $f(x,y)$

HOW TO SOLVE HOMOGENEOUS DIFF. EQUATIONS?

- (1) If a differential equation $y' = f(x,y)$ is homogeneous, then it means that it can be written as

$$y' = F(y/x)$$

- (2) Define a new variable $u = y/x$, $y = ux$.

- (3) $y' = u + xu'$ substitute it into $y' = F(u)$. Get

$$u + xu' = F(u)$$

- (4) $xu' = F(u) - u$ $\frac{du}{dx} = \frac{F(u) - u}{x}$

$$\frac{du}{F(u) - u} = \frac{dx}{x} \rightarrow \text{Solve this separate diff equation.}$$

Ex 1 $(x^2 + 2y^2)dx = xy dy$

$x^2 dx + 2y^2 dx = xy dy$ NOT SEPARABLE

$\frac{dy}{dx} = \frac{x^2 + 2y^2}{xy} = \frac{x}{y} + \frac{2y}{x}$ $f(x, y) = \frac{x^2 x^2 + 2x^2 y^2}{xxy} = \frac{x^2 + 2y^2}{xy} = f(x, y)$ Hom

$\frac{y}{x} = u \Rightarrow y = ux \quad y' = u'x + u$

$\frac{dy}{dx} = \frac{x}{y} + 2\frac{y}{x} \Rightarrow u'x + u = \frac{1}{u} + 2u \quad u'x = \frac{1}{u} + u$

$\frac{du}{dx} \cdot x = \frac{1}{u} + u \quad \frac{du}{dx} = \frac{1/u + u}{x} \rightarrow \text{separable}$

$\int \frac{u}{1+u^2} du = \int \frac{dx}{x} \Rightarrow \int \frac{du}{2u} = \int \frac{dx}{x}$

$1+u^2 = v$

$2u du = dv \Rightarrow u du = dv/2$

$\frac{1}{2} \ln|v| = \ln|x| + \ln C$

$\frac{1}{2} \ln|1+u^2| = \ln(C \cdot |x|)$

$(1+u^2)^{1/2} = C \cdot |x| \rightarrow \text{sol of separable}$

Back to original variables $|1 + (\frac{y}{x})^2|^{1/2} = C|x| \rightarrow \text{sol of Hom.}$

Check for extra solutions

$\frac{du}{dx} = \left(\frac{u^2+1}{u} \right) \cdot x$
 $q(u)$

$q(u) = \frac{u^2+1}{u} \neq 0$ NO SOLUTIONS.

$$(2) \quad x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + x$$

$$\sin\left(\frac{y}{x}\right) \frac{dy}{dx} = \frac{y}{x} \sin\left(\frac{y}{x}\right) + 1$$

$$\frac{y}{x} = u \quad y = ux$$

$$y' = u + u'x$$

$$\sin(u) \cdot (u'x + u) = u \sin(u) + 1$$

$$u'x \cdot \sin(u) + \cancel{u \sin(u)} = \cancel{u \sin(u)} + 1$$

$$u'x \cdot \sin(u) = 1 \Rightarrow \frac{du}{dx} \cdot x = \frac{1}{\sin u} \quad \int \sin u \, du = \int \frac{dx}{x}$$

separable diff.

$$-\cos u = \ln|x| + \ln C$$

$$-\cos u = \ln C|x| \quad C|x| = e^{-\cos u}$$

→ sol of separable diff.

$$C|x| = e^{-\cos(y/x)} \rightarrow \text{sol of Hom. diff.}$$

Find extra solutions

$$\frac{du}{dx} = \frac{1}{\sin u} \cdot \frac{1}{x} \quad q(u) \neq 0$$

no solution.

$$(3) \quad \frac{dy}{dx} = \frac{2xy e^{(x/y)^2}}{y^2 + y^2 e^{(x/y)^2} + 2x^2 e^{(x/y)^2}}$$

look for REVERSE

$$\frac{dx}{dy} = \frac{y^2 + y^2 e^{(x/y)^2} + 2x^2 e^{(x/y)^2}}{2xy e^{(x/y)^2}} = \frac{y}{2x e^{(x/y)^2}} + \frac{y}{2x} + \frac{x}{y}$$

$$\frac{x}{y} = u \quad x = uy \quad \frac{dx}{dy} = u'y + u$$

$$u'y + u = \frac{1}{2ue^{u^2}} + \frac{1}{2u} + u \quad \frac{du}{dy} \cdot y = \frac{1+e^{u^2}}{2ue^{u^2}} \rightarrow \text{separable}$$

$$\int \frac{2ue^{u^2}}{1+e^{u^2}} du = \int \frac{dy}{y} \Rightarrow \ln|1+e^{u^2}| = \ln|y| + \ln C$$

$$|1+e^{u^2}| = C|y| \rightarrow \text{sol of sep.}$$

$$1+e^{u^2} = v \quad 2ue^{u^2} du = dv \quad (20)$$

Sol of Hom. $\Rightarrow \overbrace{1 + e^{(x/y)^2}} = Cy$

Extra solution $\frac{dy}{dx} = \underbrace{\frac{1 + e^{y^2}}{2ye^{y^2}}}_{Q(y)} \cdot \frac{1}{y}$ $Q(y) \neq 0$ No extra solution

EXERCISES

① $(3x + 2y)dx + 2xydy = 0$ $2y dx \Rightarrow$ violates sep. eq.

$\frac{dy}{dx} = -\frac{3x+2y}{2x} = -\frac{3}{2} - \frac{y}{x}$ $y/x = v \Rightarrow y = vx$
 $\underbrace{\quad}_{F(y/x)} \quad y' = v'x + v$

$v'x + v = -\frac{3}{2} - v \quad v'x = -\frac{3}{2} - 2v \quad \frac{dv}{dx} \cdot x = -\frac{3-4v}{2} \rightarrow \text{sep.}$

$\int \frac{2}{3+4v} dv = -\int \frac{dx}{x} \Rightarrow \frac{1}{2} \ln|3+4v| = -\ln|x| + \ln C$

$3+4v = v$
 $4dv = dv$

$|3+4v|^{1/2} = \frac{C}{|x|} \Rightarrow \text{Sol. of sep. diff.}$

$|3+4y/x|^{1/2} = \frac{C}{|x|} \Rightarrow \text{sol of hom.}$

Extra sol $\frac{dy}{dx} = \frac{-3-4v}{2} \cdot \frac{1}{x}$ $Q(v) = 0$

$v = -3/4$ can be found from

$C=0$
 No need for extra sol.

⊛ Consider we have found an extra solution $v=L$ for sep. equation, then $v = \frac{y}{x} = L \quad y = Lx$ is extra sol. of hom. diff. equation.

② $(x^2 + by^2) dx - 4xy dy = 0$ $y(1) = 1$

$$\frac{dy}{dx} = \frac{x^2 + by^2}{4xy} = \frac{x^2}{4xy} + \frac{by^2}{4xy} = F\left(\frac{y}{x}\right) \rightarrow \text{Homog.}$$

$$y/x = u \Rightarrow y = u'x + u$$

$$u'x + u = \frac{1}{4u} + \frac{3}{2}u \Rightarrow u'x = \frac{1}{4u} + \frac{3}{2}u \Rightarrow \frac{du}{dx} = \underbrace{\left(\frac{2u^2 + 1}{4u}\right)}_{f(u,x)} \cdot \frac{1}{x}$$

$$\frac{4u}{2u^2 + 1} du = \frac{dx}{x} \Rightarrow \ln|2u^2 + 1| = \ln|x| + \ln C$$

$$2u^2 + 1 = C|x| \rightarrow \text{sol of sep.}$$

$$2\left(\frac{y}{x}\right)^2 + 1 = C|x| \rightarrow \text{sol of homog.}$$

No extra solution from $q(u) = 0$

③ $\frac{dy}{dx} = \frac{y}{x + \sqrt{xy}}$ $x > 0$

$$f(x,y) = \frac{y}{x + \sqrt{xy}} \Rightarrow f(\alpha x, \alpha y) = \frac{\alpha y}{\alpha x + \sqrt{\alpha x \cdot \alpha y}} = \frac{\alpha y}{\alpha(x + \sqrt{xy})} = \frac{y}{x + \sqrt{xy}} = f(x,y)$$

$$\frac{y}{x} = u \quad y = ux \quad y' = u'x + u$$

$$u'x + u = \frac{ux}{x + \sqrt{x} \sqrt{ux}} \Rightarrow u'x + u = \frac{u}{1 + \sqrt{u}}$$

$$u'x = \frac{u}{1 + \sqrt{u}} - u = \frac{u - u - u^{3/2}}{1 + \sqrt{u}} \Rightarrow \frac{du}{dx} \cdot x = \underbrace{\left(\frac{-u^{3/2}}{1 + u^{1/2}}\right)}_{q(u)}$$

$$\int \frac{1 + u^{1/2}}{u^{3/2}} du = - \int \frac{dx}{x} \rightarrow \text{Sep.}$$

$$\int \left(\frac{1}{u^{3/2}} + \frac{1}{u}\right) du = - \int \frac{dx}{x} \quad -2u^{-1/2} + \ln|u| = -\ln x - \ln C$$

General sol. of separable.

Extra sol of Sep. diff $q(u) = 0$ $u = 0$

$$-2\left(\frac{y}{x}\right)^{-1/2} + \ln\left|\frac{y}{x}\right| = -\ln x - \ln C$$

General sol of homog. Extra sol of homog $u = \frac{y}{x} = 0 \Rightarrow y = 0$

$$(4) (3x^2 - y^2) dx - 2xy dy = 0$$

$$(3x^2 - y^2) dx - 2xy dy = 0$$

$$\frac{3x^2 - y^2}{2xy} = \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{3x^2 - y^2}{2xy}$$

$$\frac{dy}{dx} = \frac{3x}{2y} - \frac{y}{2x} = F(y/x) \quad \frac{y}{x} = v \Rightarrow y' = v'x + v$$

$$v'x + v = \frac{3}{2v} - \frac{v}{2} \Rightarrow v'x = \frac{3}{2v} - \frac{3v}{2} = \frac{3 - 3v^2}{2v}$$

$$\frac{dv}{dx} \cdot x = \frac{3 - 3v^2}{2v} \quad \frac{2v}{3 - 3v^2} dv = \frac{dx}{x}$$

$$\int \frac{2v}{1 - v^2} dv = \int 3 \frac{dx}{x}$$

sol of sep. $\ln C - \ln |1 - v^2| = 3 \ln |x|$

$$\ln C = \ln |1 - v^2| + 3 \ln |x|$$

$$C = |1 - v^2| \cdot |x|^3 \rightarrow \text{sol of sep. } (*)$$

Extr. sol $q(v) = \frac{1 - v^2}{2v}$ $q(v) = 1 - v^2 = 0 \quad v = \pm 1$

Can be obtained from
 (*) with $C = 0$ no
 extr. sol.

Sol of Hom. $C = |1 - y/x^2| \cdot |x|^3 \rightarrow \text{sol of hom.}$

(5) $xyy' + 4x^2 + y^2 = 0$ $y(2) = -7$.

$xy \frac{dy}{dx} = -4x^2 - y^2$ $\frac{dy}{dx} = -\frac{4x}{y} - \frac{y}{x} = F(y/x) \rightarrow \text{hom.}$

$\frac{y}{x} = v$ $y = vx$ $y' = v'x + v$

$v'x + v = -\frac{4}{v} - v$ $v'x = -\frac{4+2v^2}{v}$ $\frac{dv}{dx} \cdot x = -\frac{4+2v^2}{v}$

$\frac{v \cdot dv}{4+2v^2} = -\frac{dx}{x} \Rightarrow \frac{1}{4} \ln |4+2v^2| = -\ln|x| + \ln C$

General Sol of sep.

General sol of hom. $\frac{1}{4} \ln |4+2v^2| = -\ln|x| + \ln C$

$\frac{1}{4} \ln |4+2(\frac{y}{x})^2| = -\ln|x| + \ln C$

$(4+2(\frac{y}{x})^2)^{1/4} = \frac{C}{|x|}$

Particular sol of hom. $(4+2(-\frac{7}{2})^2)^{1/4} = \frac{C}{2}$ $C = (\frac{7}{2})^{1/4}$

$(4+2(\frac{y}{x})^2)^{1/4} = \frac{2 \cdot (\frac{7}{2})^{1/4}}{|x|}$

(6) $xy' = y(\ln x - \ln y)$ $y(1) = 4$ $x > 0$

$\frac{dy}{dx} = \frac{y}{x} \ln(\frac{x}{y}) = F(y/x)$ $\frac{y}{x} = v \Rightarrow y = vx$ $y' = v'x + v$

$v'x + v = v \ln v^{-1}$ $v'x + v = -v \ln v$ $v'x = -v \ln v - v$

$\frac{dv}{dx} \cdot x = -v(\ln v + 1)$

$\int \frac{dv}{v(\ln v + 1)} = -\int \frac{dx}{x} \rightarrow \text{sep.}$

$\ln v + 1 = u$

$\frac{du}{v} = dv$

$\ln|v| = -\ln|x| + \ln C$

$|v| = \frac{C}{|x|}$

$(\ln v + 1) = \frac{C}{|x|} \rightarrow \text{General sol of sep.}$

(24)

$$\left| \ln \frac{y}{x} + 1 \right| = \frac{c}{|x|} \rightarrow \text{general sol of hom.}$$

$$\left| \ln \frac{y}{x} + 1 \right| = c \quad c = \ln 4 + 1$$

Particular sol. $\left| \ln \frac{y}{x} + 1 \right| = \frac{\ln 4 + 1}{|x|}$

⑦ $y' = 3 \cdot \frac{x+y}{x} \quad y(4) = 0$

$$y' = 3 \left(1 + \frac{y}{x} \right) \quad y = vx \quad y' = v + v'x$$

$$v + xv' = 3(1+v) \quad v'x = 3 + 2v \quad \frac{dv}{2v+3} = \frac{dx}{x}$$

$$\frac{1}{2} \ln |2v+3| = \ln |x| + \ln C$$

$$(2v+3)^{1/2} = C|x| \rightarrow \text{general sol of sep.}$$

$$(2 \cdot y/x + 3)^{1/2} = C|x| \rightarrow \text{general sol of hom.}$$

$$(2 \cdot \frac{0}{1} + 3)^{1/2} = C \cdot 1 \quad C = \sqrt{3}$$

$$\left| 2 \frac{y}{x} + 3 \right|^{1/2} = |\sqrt{3}x| \rightarrow \text{Particular sol of Hom.}$$

⑧ $y' = \frac{y^2 + x^2}{yx} \quad \left(\frac{dy}{dx} \right) = \frac{y}{x} + \frac{x}{y} = y/x + 1/y/x = F(y/x)$

$$y(4) = 2$$

$$y = vx$$

$$y' = v + xv'$$

$$v + xv' = v + \frac{1}{v} \quad xv' = 1/v \text{ sep.}$$

$$x \cdot \frac{dv}{dx} = \frac{1}{v}$$

$$v dv = \frac{dx}{x}$$

$$\frac{v^2}{2} + C = \ln |x| \rightarrow \text{General sol. of sep.}$$

$$\left(\frac{y}{x} \right)^2 + C = \ln |x| \rightarrow \text{General sol of hom.}$$

$$\left(\frac{2}{1} \right)^2 + C = \ln |1| \quad C = 2 \quad \left(\frac{y}{x} \right)^2 + 2 = \ln |x| \rightarrow \text{Particular sol of hom.}$$